

Advanced Topics in Monetary Economics with Heterogenous Agents

— Problem Set 1, Spring 2021 —

Due: Friday, Feb 5, 11:59pm on Canvas

Important guidelines.

1. Please only submit this problem set if you are *not auditing* this course and are *officially enrolled* for credit. Each problem set counts for 20% of your final grade.
2. Make sure to *submit your code*. The best way to solve our problem sets is using Jupyter Notebooks. We *strongly suggest using Python* from the beginning since this course will progressively build on existing Python modules in future problem sets. You can ask your classmates coding questions in Slack, Martin in OH, and review QuantEcon's programming tutorial.¹
3. The problem sets for this class are meant to give you exposure to the actual, messy research process. This means: If a question is incomplete or missing information, please go ahead and make an assumption that you deem reasonable. Be sure to *state your assumptions and submit your full derivations* in your solution. Many problems could have multiple solutions.

1 Tractable “precautionary saving” model

Consider the following perpetual-youth economy in continuous time. There is a mass 1 of agents. Each instant, a flow of δ agents are born and die. The agents that are born begin their life with exogenous wealth a_0 . They have a discount factor of ρ and face a constant interest rate $r < \rho + \delta$. They do not earn any income and face a borrowing constraint $\underline{a}$. They have log utility. Thus, agents maximize

$$\max \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} s_t \log c_t dt \right]$$

subject to

$$\dot{a}_t = ra_t - c_t \tag{1}$$

$$a_t \geq \underline{a}$$

Here, $s_t = 1$ as long as agents are alive, and 0 otherwise.

¹<https://python-programming.quantecon.org/intro.html>

1. Show that this problem is equivalent to the deterministic consumption-savings problem maximizing utility

$$\int_0^{\infty} e^{-(\rho+\delta)t} \log c_t dt$$

2. Derive the Euler equation for the agent using your preferred dynamic optimization method.
3. Postulating a consumption function $c(a; \underline{a})$. Henceforth, we also denote $c(a; \underline{a})$ by $\mathbf{c}(a)$, and its inverse by $\mathbf{a}(c)$. Derive a single ODE for $\mathbf{c}(a) \equiv c(a; \underline{a})$ based on the Euler equation and (1). What is the boundary condition? [Hint: this question does NOT ask you to guess a parametric functional form for $c(a; \underline{a})$ nor to solve the ODE just yet, simply to write down a simple ODE as a function of a]
4. Prove that, $c(a; \underline{a})$ is homogeneous of degree 1 in $(a, \underline{a})$.

After solving questions 1-4, you are ready now to solve the ODE.

5. Transform the ODE, which is stated in terms of $\mathbf{c}(a)$, into one stated in terms of $\mathbf{a}(c)$. [Hint: This means, only $\mathbf{a}(c)$, $\mathbf{a}'(c)$ appear, not $\mathbf{c}(a)$.] You should arrive at

$$(\rho + \delta - r) \cdot c \cdot \mathbf{a}'(c) + r \cdot \mathbf{a}(c) - c = 0 \quad (2)$$

What is the boundary condition?

6. Solve the ODE, subject to the correct boundary condition. [Hint: First solve the homogeneous part of the ODE, excluding the last term in (2). Then find a particular solution of (2). You should find $\mathbf{a}(c) = \underline{a} \frac{1}{k+1} \left(\frac{c}{r\underline{a}} \right)^{-k} + \underline{a} \frac{k}{k+1} \frac{c}{r\underline{a}}$.]
7. Show that $\mathbf{a}(c)$ is convex, and thus, that the consumption function $\mathbf{c}(a)$ is concave.
8. For this question only, assume $r = (\rho + \delta) / 2$. Compute $\mathbf{c}(a)$ explicitly.
9. In the case with general r, ρ , can you use your solution for $\mathbf{a}(c)$ to prove that for a close to $\underline{a}$, $\mathbf{c}(a) \approx r\underline{a} + \text{const} \cdot \sqrt{a - \underline{a}}$? Describe the behavior of MPCs $\mathbf{c}'(a)$ as $a \rightarrow \underline{a}$. Why does the MPC not converge to 1?
10. Prove that the MPC is always above r .
11. [open-ended] There is absolutely no idiosyncratic risk in this model (other than mortality risk, which just changes the effective discount rate). And yet, we found a concave consumption function. Why is it concave nevertheless? Can you devise a test that one could apply to a standard precautionary saving model with idiosyncratic risk to figure out to what extent the concavity of the consumption function is driven by idiosyncratic risk at all vs. the mechanisms at work in this model?

2 Solving the Huggett model

In this question, we solve a Huggett model, step by step. There are two way to do this question. The first way involves you writing every line of code yourself, only looking at any code we might give you (or find online) for inspiration. The second way involves you using any code we might give you (or find online). Both are graded equally. The first way is encouraged for students with prior coding experience or prior experience solving heterogenous-agent models. We strongly suggest solving this problem in a Jupyter Notebook and using Python from the beginning since this course will progressively build on existing Python modules that have been tailored to solve the models we will cover in this course.

Model. There is a mass 1 of agents in the model. Each agent solves a consumption saving problem of the form

$$\begin{aligned} \max \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_t) \right] \\ c_t + a_t \leq (1 + r_{t-1})a_{t-1} + s_t Y_t \\ a_t \geq \underline{a} \end{aligned}$$

where $\log s_t$ follows an AR(1) process,

$$\log s_t = \rho \log s_{t-1} + \epsilon_t$$

where ρ is the persistence and $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$ iid. $\underline{a}$ is the borrowing constraint (typically this is negative; when $\underline{a} = 0$, agents cannot borrow).

1. **Recursive formulation.** Write down the Bellman equation associated with this program in terms of $V_t(\mathbf{x}_t) = V(\mathbf{x}_t; \mathbf{X}_t)$, the value function of an agent taking two sets of inputs, the idiosyncratic state vector $\mathbf{x}_t$ and the aggregate state vector $\mathbf{X}_t$, which you need to specify. [Hint: think about the dimension of these two state vectors from the point of view of the agent].
2. **Optimality conditions.** Write down the F.O.C. and the envelope condition. [Note: having these handy will be very important in the computation of the model].

For now, consider the steady state first.

3. **Steady state.** Show how the economy being in the steady state simplifies the Bellman equation [Hint: think about what the aggregate state $\mathbf{X}_t$ you specified above collapses to in the steady state]

Our baseline quarterly calibration will be $Y = 1$, $r = 0.0025$, $\beta = 1 - 0.02$, $\rho = 0.975$, $\sigma_\epsilon = 0.7$, $u(c) = \log c$, $\underline{a} = -1$.

4. **The grid.** Set up the grids for the two individual state variables in the model in $\mathbf{x}_t$.

- i. Discretize the income process using the Rouwenhorst method to arrive at a finite dimensional Markov chain with transition matrix Π , independent of aggregates. Pick $n_s = 7$ grid points as your default.
 - ii. Set a grid for wealth that starts at $\underline{a}$ and goes up to some upper bound a^{max} . Pick $a^{max} = 100$ and $n_a = 500$ grid points as your default.
5. **Backward iteration.** Write a backward iteration function: [this method is a version of Chris Carroll's *Endogenous Grid-point Method*]

- i. The function takes as given (as an input) a *guess* for the marginal value tomorrow $\frac{\partial V_{t+1}}{\partial a_t}(s_{t+1}, a_t)^0$, on a $n^s \times n^a$ array.
- ii. It then applies the first order condition you stated above to derive an expression for $c_t^{next} = c_t(s_t, a_t)$ as function of current s_t but *next period's* asset position a_t . c_t^{next} is also an $n^s \times n^a$ array.
- iii. How do we arrive at a policy function $c_t(s_t, a_{t-1})$ as function of the *current* asset position a_{t-1} ? To do so, we use the budget constraint

$$c_t^{next} + a_t = (1 + r_{t-1}) a_{t-1} + s_t Y_t$$

For every given s_t , the left hand side $LHS(a_t)$ is a function of next period's assets a_t . We know what $LHS(a_t)$ is on each grid point for a_t . Similarly, the right hand side $RHS(a_{t-1})$ only depends on a_{t-1} . Use linear interpolation to find the approximate grid point next period, a_t , corresponding to any given a_{t-1} today.

- iv. For some values of a_{t-1} , the associated a_t may lie below $\underline{a}$. Please make sure to replace a_t so the borrowing constraint is never violated.
 - v. Following these steps for every s_t gives us the asset policy $a_t = \mathbf{a}_t^*(s_t, a_{t-1})$. Compute the consumption policy on the current grid from the budget constraint
 - vi. From this and the envelope condition you derived above, we can compute the new marginal value on the current income and asset grid
 - vii. You're done! When doing policy function iteration now, all you will have to check is whether your new guess $\frac{\partial V_t}{\partial a_{t-1}}(s_t, a_{t-1})^1$ is close to old one.
6. **Policy functions.** Apply the backward iteration function iteratively to find the steady state policies, when $r_t = r$ and $Y_t = Y$ are given. Plot the consumption and asset policies over a_{t-1} for several income states.
7. **Forward iteration.** Next, we write a forward iteration function: [This method is also known as Young's *histogram* method.]

- i. The function takes as given a guess for the distribution $D_{t-1}(s_t, a_{t-1})$ (here meant to be a pdf, not a cdf). Again, this is an $n^s \times n^a$ array. Our goal is to find $D_t(s_{t+1}, a_t)$.

ii. We first construct the distribution over current income states and next period's asset states (today's savings or borrowing position), denoted by $\tilde{D}_{t-1}(s_t, a_t)$. To do so, we loop over each current asset grid point a_{t-1}

i. Let the asset policy for this grid point, $\mathbf{a}_t^*(s_t, a_{t-1})$, lie between two arbitrary next period asset grid points a_t^0 and a_t^1 . Thus, it can be expressed as a linear combination

$$\mathbf{a}_t^*(s_t, a_{t-1}) = \lambda a_t^0 + (1 - \lambda) a_t^1$$

ii. We then assign mass $\lambda D_{t-1}(s_t, a_{t-1})$ to $\tilde{D}_{t-1}(s_t, a_t^0)$ and $(1 - \lambda) D_{t-1}(s_t, a_{t-1})$ to $\tilde{D}_{t-1}(s_t, a_t^1)$, much like when plotting a distribution's histogram, hence the name

iii. Finally, we need to iterate on the exogenous state, and apply the transpose of the transition matrix Π of s to $\tilde{D}_{t-1}(s_t, a_t)$ to obtain $D_t(s_{t+1}, a_t)$.

8. **Aggregation under a stationary distribution.** Apply your forward iteration function iteratively using the steady state policy $\mathbf{a}^*(s_t, a_{t-1})$ to find the steady state or stationary distribution $\mathbf{D}(s_t, a_{t-1})$. Show it in a histogram. Compute aggregate asset holdings A and aggregate consumption C implied by your steady state policies and the distribution of agents for the default parameter values.
9. **Aggregate savings supply.** Next, we compute the saving supply curve. Compute the steady state for interest rates in some interval, e.g. $r \in [-0.02, 0.02]$. Create one plot showing steady state asset holdings $A(r)$, and one plot showing steady state consumption $C(r)$.
10. **General equilibrium steady state.** In the Huggett economy there's neither a government nor proper firm to demand net households savings, and thus, assets must be in zero net supply. Compute and report the general equilibrium steady state (i.e. the interest rate at which asset holdings are zero, $A = 0$) and report how long your code takes to find the steady state. The best way to numerically find the unique interest rate r for which $A = 0$ is to use Brent's method. An alternative way that is sometimes being propose is to solve for the steady state r using *goods* market clearing $C = Y$ rather than *asset* market clearing. Using your answer to the previous question, comment on why this may not be a great idea.
11. **General equilibrium transitional dynamics.** [This requires you to use a *shooting* method]. In our final part of this problem, we compute the GE transitional dynamics to an exogenous Y_t shock. To do so, assume the economy starts in the steady state, and let Y_t follow a pre-specified path

$$Y_t = Y + dY \cdot \eta^t$$

where Y is the steady state level, $Y = 1$, dY is the magnitude of the shock, and $\eta \in (0, 1)$ is the decay factor of the shock. For this exercise, assume $dY = 0.1$ and $\eta = 0.90$. To simulate the transitional dynamics, we are looking for the path of interest rates $\{r_t\}$ such that asset

market clearing holds in each period, that is, $A_t = 0$ for all t . The following is a standard shooting algorithm to do this:

- i. Guess a candidate *path* for interest rates $\{r_t\}$.
- ii. Given the path, solve for policy functions by iterating backwards, just like before. This time, assume income in period t is Y_t , not the steady state Y . Start at some large time, e.g. $t = T = 100$, and assume terminal marginal utility $\frac{\partial V_{T+1}}{\partial a_T}$ is the steady state one. You should arrive at a sequence of policy functions $\mathbf{c}_t^*, \mathbf{a}_t^*$.
- iii. Use the policy functions to iterate forward, starting with the steady state distribution D_{-1} at date $t = 0$. You should arrive at a sequence of distributions $\{D_t\}$.
- iv. Compute a candidate sequence of total assets $\{A_t\}$ as the average asset position when the distribution is D_t in each period.
- v. Check whether each element in $\{A_t\}$ is close to zero. If not, adjust $\{r_t\}$ appropriately thinking about the economics of the problem. [*Hint: think about how each r_t needs to change if A_t is positive or negative for asset markets to clear*]. Plot the impulse response function of $\{r_t\}$ to the shock and report how long your code takes to compute it.

Congratulations! You have now solved your first heterogeneous agent model with perfect-foresight aggregate dynamics and are well equipped for this course.